

BEVISONNEERS

THE MERCEDES-BENZ FELLOWSHIP

Planet-positive Project Portfolio
BioViL

PANKAJ SHARMA

The Vision

Introduction

The beginnings and future of this project

The idea behind bringing knowledge and awareness to the idea of bioenergy solution for communities developed when I first started a tree plantation initiative in our village and heard a question as, "Why do we need to plant trees, when we already have a lot around us?". That was the moment when I realised that mere implementation will not be sufficient, if there isn't awareness, knowledge and a sense of belonging into it, which in turn will help develop the right mind set needed to effectively bring change.

About me

My ***personal purpose*** is to help communities, from the rural parts, to primarily make them aware of sustainability, its necessity, and to better utilize the abundance of biomass resources (animal and household waste) so that they are equipped with the tools and solutions that are more cleaner and greener.

I/we aspire to work with resilience, integrity and sustainable habits, because it helps me to find hope, and that hope can then turns into a vision to build a just, fair, and equitable planet.

The System

The central issue

- Under-utilization of the available biomass resources and burning of firewood for different uses in rural communities.

Key variables

- Awareness and Education, Deforestation, Green Solutions, Affordable energy alternatives.

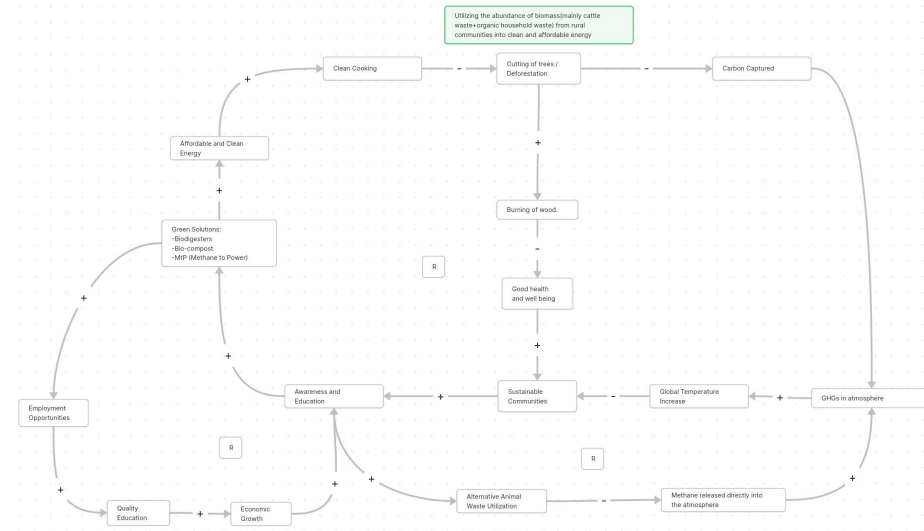
The strongest relationships (+ and -)

- Economic growth -> Awareness and Education(+)
- Clean Cooking -> Deforestation(-)

Key reinforcing patterns (feedback loops)

- Awareness -> Green Solutions -> Employment -> Quality Education -> Economic Upliftment -> Awareness

Reflection: The Problem is occurring due to easy access to the forest resources, lack of alternatives and primarily no awareness or knowledge on this domain.



The Root Causes

Direct and indirect drivers, beliefs and values

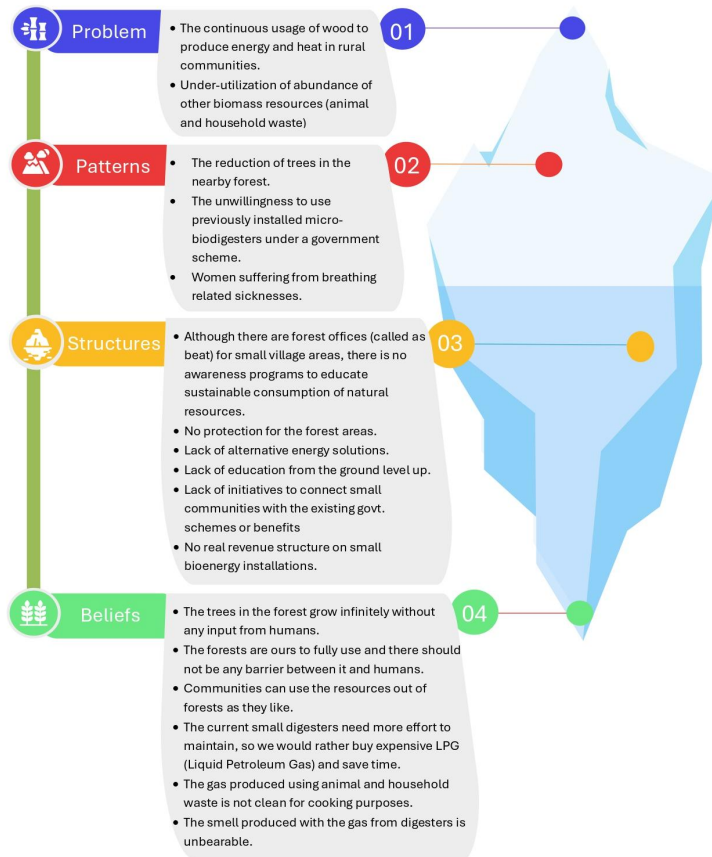
Structures contributing to [problem]:

- No awareness programs to educate sustainable consumption of natural resources.
- No protection for the forest areas.
- Lack of alternative energy solutions.
- No revenue structure on small bioenergy installations.
- Lack of education on sustainability from primary level.

Beliefs contributing to [problem]:

- The trees in the forest grow infinitely.
- The forest resources are ours to fully use.
- Communities can use the resources out of forests as they like.
- The bio-digesters need more effort and time to operate.
- The gas produced using animal waste cannot be used for cooking purposes (unhygienic).
- The smell of the gas produced from the digesters is unbearable.

BioViI – The Iceberg Model



The People

User Insights

People talked to:

- To a villager, with a cow farm producing about 200 kg of animal waste, which goes completely unused

Methods used:

- Interview (Online)

Needs and pain points identified:

- Capital needed, no knowledge, no motivation

Current solution available:

- To use some waste as a natural compost for growing vegetables

Desires & bright spots found:

- Willingness to construct a test plant without any charges, and cooking fuel(wood from trees) for animal food can be substituted by biogas.

Purpose-driven Problem Statement

Weaving all the findings
together

In summary

I want to address the problem of **deforestation, clean cooking, and underutilization of biomass resources among rural communities.**

I have identified **education** turning into **willingness to implement green affordable solutions** which in turn results in **Economic Upliftment**, which relates to deeper underlying beliefs as: **trees grow infinitely, we are free to use the forest resources as we want, etc.,** revealing the root cause as **lack of awareness among people.**

The **community** I serve come from rural regions of India, taking into account their **lacking access to clean cooking, and reliable and affordable electricity.**

I aspire to **help people from rural communities find hope through innovative, sustainable, and engaging solutions**, while living up to **integrity, gratefulness, resilience and sustainable habits.**